



A novel machine learning framework for designing high-performance catalysts for production of clean liquid fuels through Fischer-Tropsch synthesis

Zahra Teimouri^a, Nicolas Abatzoglou^b, Ajay K. Dalai^{a,*}

^a Catalysis and Chemical Reaction Engineering Laboratory, Department of Chemical and Biological Engineering, University of Saskatchewan, Saskatoon, SK, S7N 5A9, Canada

^b Department of Chemical and Biotechnological Engineering, Université de Sherbrooke, Sherbrooke, QC, J1K 2R1, Canada

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ABSTRACT

Fischer-Tropsch synthesis (FTS) attracts great interest as a sustainable route for production of sustainable transportation fuels. Catalyst design and operational conditions tune the selectivity of FTS to liquid fuels, which can be predicted with machine learning models (ML). Herein, a comprehensive dataset including 27 key input features related to the catalyst structure (with the focus on carbon supports), preparation, activation, and FTS operating conditions were investigated to predict the CO conversion and C₅₊ selectivity using three ML algorithms. Feature engineering and selection were implemented to identify the significant catalyst formulation descriptors. Principal component analysis was used to explore the information space and decrease the dimension of the dataset before ML prediction. In addition to the well-known effects of the operational conditions on FTS, roles of physico-chemical properties of the carbon materials were also extensively analyzed. Random forest (RF) including 4 principal components, indicated the highest accuracy for prediction of the CO conversion and C₅₊ selectivity with R² of 0.91 and 0.97, respectively. The proposed framework can provide a reliable strategy in designing efficient carbon supported catalysts for FTS and guide experiments by identifying the key descriptors in the catalyst formulation and operating conditions to enhance the selectivity of liquid fuels.

1. Introduction

The increased energy demand and global warming have sparked a heightened focus on developing renewable and sustainable energy resources [1]. Fischer-Tropsch synthesis (FTS) is a catalytic process that converts syngas (CO + H₂) to light olefins, liquid fuels, and value-added chemicals, that has received a renewed attention during the last decade [2]. Syngas can be derived through sustainable processes such as biomass gasification [3]. Light olefins, a classification of the FT products, are considered the building blocks for production of the valuable chemicals. On the other hand, liquid hydrocarbons produced through FTS, owing to approximately zero amounts of sulfur and aromatics are assumed as clean transportation fuels [4].

The selectivity of FTS process highly depends on the catalyst, reactor designs, and operational conditions. Co and Fe catalysts are among the most common active phases for hydrogenation of the syngas in FTS. Catalyst design consists of catalyst preparation techniques, support

design, promoter selection, calcination, and reduction conditions. Incorporation of the several reaction steps in FTS, makes the products prediction challenging. Modeling approaches can provide an alternative method to resolve the present challenges in defining a reliable reaction mechanism and products distribution for FTS by predicting the optimal catalyst structure and operational conditions [5,6].

Machine learning (ML) techniques have been widely used to uncover the black box of the catalytic reactions with controversial reaction mechanisms such as FTS [7]. Garona et al. [5] used artificial neural network (ANN) to predict the FTS performance of the catalysts with different formulations. The active phase, support, promoter types, and FTS operational conditions were used as the input data to train and test the capability of ANN model to predict the FTS performance in terms of CO conversion, light olefins selectivity, and CO₂ yield. According to the ANN model and sensitivity analysis results, FeK/TiO₂ and CoMn/TiO₂ catalysts at operating temperature of 500 °C led to the highest light olefins selectivity. The provided model did not consider the key features

* Corresponding author.

E-mail address: akd983@mail.usask.ca (A.K. Dalai).

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of the catalyst formulation such as surface area, particle size, etc. In another work, Khorashadizadeh et al. [8] studied a hybrid ANN-RSM approach to analyze the effects of process conditions on the products distribution of the CoRe/Al₂O₃ catalyst using a small dataset. Although the provided hybrid model successfully predicted the products selectivity over various range of operational conditions, but it lacks the generalization capability for different catalyst designs. The textural and structural properties of the catalyst and catalyst support such as specific surface area, particle size, pore size, and surface chemistry play a significant role in tuning the products selectivity in FTS [9]. In FTS, like other heterogeneous catalytic reactions, the active phases are dispersed on a surface of a support material, which not only facilitates the exertion of high active metal loadings, but also amends the reaction mechanism and products selectivity.

Metal oxides and carbonaceous supports comprise most of the employed supports for FTS catalysts. While the former group of the catalyst supports usually lead to strong metal-support interactions, which impedes the catalyst reduction; the latter group adjusts the metal-support interactions [10]. Carbon-based materials with versatile surface chemistry, tunable textural features, high thermal stability, and electrical conductivity play a leading role in anchoring the FTS active sites. The metal-support interaction alters the electron density around the active metal affecting the metal-ligand bond strength in FTS [11]. On the other hand, the acidic or basic nature of the carbon support highly contributes to the products distribution. While the acidic functional groups on the surface of the support promote isomerization and cracking reactions; the support material with basic functional groups can improve the selectivity of the heavier hydrocarbons [12]. The different classifications of the carbon materials as catalyst support are visualized in Fig. 1.

Studying the effects of catalyst design as well as operational conditions sheds light on the reaction mechanism of FTS and can facilitate the screening process of the different catalyst structures. Data-driven techniques can provide a platform to build constructive models with high-dimensional data to predict the FTS performance of different catalyst structures.

Although, some studies in the literature investigated the products distribution of the FTS with ML approaches, but a sound model accounting for the simultaneous effects of the catalyst structure and operational conditions (catalyst synthesis, reduction, and FTS process) with a focus on carbon-based supports is missing. Herein a comprehensive dataset including the key structural and textural features of the catalyst as well as the operating conditions were compiled and thoroughly investigated using principal component analysis (PCA).

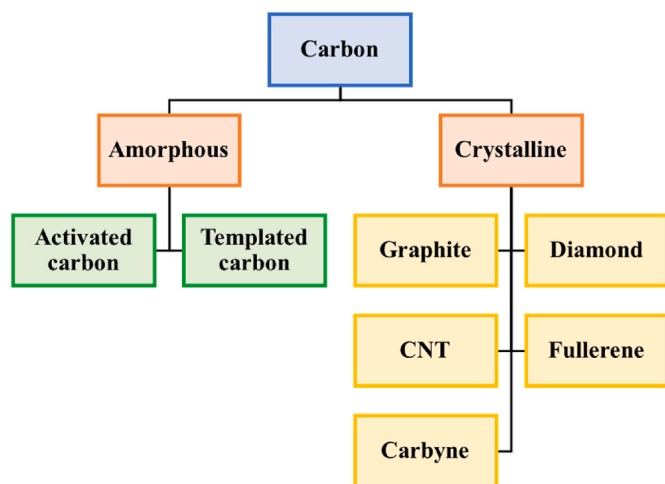


Fig. 1. Different groups of the carbon materials as catalyst support for FTS [11].

Compiling a dataset consist of both quantitative and qualitative features helps us to capture the FTS performance of the more diverse catalyst designs, preparation methods, activation, and FTS operating conditions. Three ML models (ANN, RF, and SVR) were used for prediction of the CO conversion and C₅₊ selectivity followed by Spearman correlation analysis to investigate the possible correlations between the input factors. Moreover, the impacts of the catalyst formulation and operational descriptors on the catalytic activity and selectivity were extensively scrutinized using SHAP theory. Partial dependency plots (PDPs) were also implemented to interpret the behavior of the ML models. This study provides us with a roadmap of designing efficient carbon-based catalysts for production of the non-crude oil-based transportation fuels.

2. Methodology

2.1. Data Collection

Data curation is the principal requirement for developing an effective ML algorithm and recognizing the significance of the input data prevents dealing with a flawed dataset commonly referred to as “garbage in, garbage out” [13]. The dataset in this work was collected using the experimental results from 42 recent publications (210 datasets) using various carbon structures as the catalyst supports for FTS. The compiled dataset was first statistically scrutinized and then mechanistically discussed. We classified the data into 27 different descriptors in terms of catalyst formulation, and operational conditions (Table 1 and Table 2). Due to the significance of producing sustainable liquid fuels from FTS; C₅₊ selectivity and CO conversion were set as the responses of the ML models. The entire compiled data set can be found in the supplementary material. Moreover, the details of the included features in the machine learning framework were discussed in the supplementary material (section S1). The different features of the two descriptors (catalyst formulation and operational conditions) are summarized in Tables 1 and 2

2.2. Data evaluation and model construction

To visualize the distribution of the data set and identify the potential outliers, box-plot analysis was carried out using MATLAB R2021b. The possible non-linear associations between the input variables and targets were also analyzed using Spearman correlation matrix, which was

Table 1
Descriptors of catalyst formulation.

Feature name	Range	Units
Active metal	Fe, Co, and Fe-Co	–
Support materials ^a	AC, CNT, Biochar, CNF, Chitosan, MOF, Carbon black, SMC, GNS, GO, and CEMNPs	–
Promoters	Zr, K, Na, Mg, Bi, Mo, Ru, Al, V, S, KNO ₃ , Ce, Ca, Cu, Pb, Si, Mn, Ti, Sr, KCO ₃	–
Active metal loading	0–66	weight %
Promoter loading	0–12	weight %
Average pore size of catalyst	1.04–20.5	nm
Average pore size of support	0.9–19.4	nm
Specific surface area of catalyst	2.2–1326	m ² /g
Specific surface area of support	5–1940	m ² /g
Active metal particle size	1.52–155.6	nm

^a AC: activated carbon, CNT: carbon nanotubes, CNF: carbon nanofibers, MOF: metal-organic frameworks, SMC: spherical mesoporous carbon, GNS: graphene nanosheets, GO: graphene oxide, CEMNPs: carbon encapsulated metal nanoparticles.

Table 2

Descriptors of the operational conditions for catalyst preparation, reduction, and FTS performance tests.

Feature name	Range	Units
Active metal loading method	IWT ^a , carbonization, mechanical mixing, WI ^b , hydrothermal, melting, photo-Fenton, electrostatic adsorption, microemulsion	–
Calcination time	0–12	hour
Calcination temperature	35–1000	°C
Support pre-treatment	Yes or no	–
Reduction atmosphere	H ₂ , CO, or Syngas	–
Reduction time	0–50	hour
Reduction temperature	210–450	°C
Reduction pressure	0.1–3	MPa
Reaction temperature	200–350	°C
Reaction pressure	0.8–8.62	MPa
GHSV	500–36000	(hour) ^{−1}
H ₂ /CO ratio	0.72–2	–
TOS	6–200	hour

^a Incipient wetness impregnation.

^b Wetness impregnation.

developed in Python programming platform. In the third step of the proposed ML framework, principal component analysis (PCA) was conducted to reduce the dimensionality of the experimental data set by leveraging the informative space in the data. The scikit-learn library of the Python was used for PCA analysis [14]. The steps of PCA analysis are depicted in Fig. 2.

In the next step, experimental dataset was introduced to three ML models (support vector regression (SVR), random forest (RF), and ANN) to predict the catalytic activity and selectivity as a function of catalyst design and operating conditions. Before data training and testing, standard scaler in the scikit-learn library was employed for data normalization using Python programming. The details of the ML models investigated in this work has been discussed in the supplementary material (section S.2). The prediction performances of the ML models were evaluated using R^2 (correlation coefficient) and RMSE (root mean square error), which are calculated based on the following formulas:

$$R^2 = 1 - \frac{\sum_{n=1}^N (y'_n - y_n)^2}{\sum_{n=1}^N (y'_n - \bar{y})^2} \quad (1)$$

$$RMSE = \sqrt{\frac{\sum_{n=1}^N (y'_n - y_n)^2}{N}} \quad (2)$$

where, y'_n is the predicted value of the target response from ML model, y_n

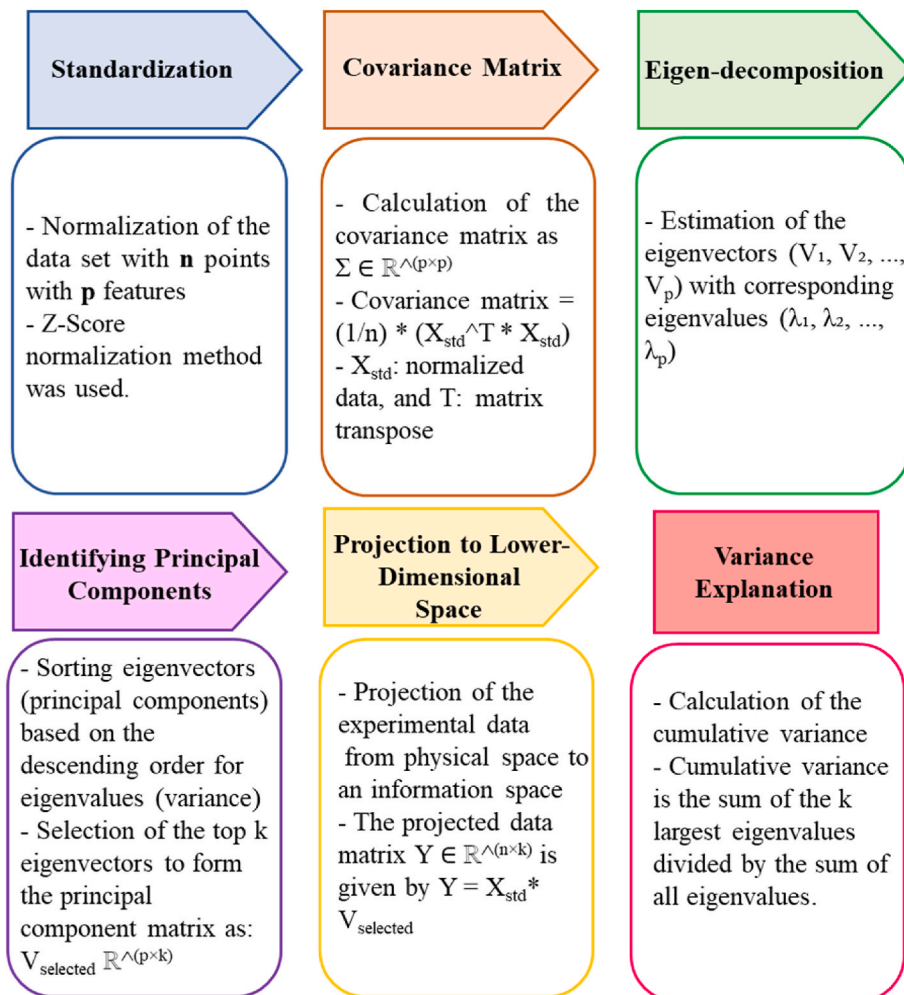


Fig. 2. Schematic of the different steps in PCA analysis.

represents the actual value of the output target; $\bar{y}$ is the average value of each target; n is the index of datapoint samples for each target, and N stands for the total number of datapoints.

In the last step of the ML framework, model interpretation tools were applied to investigate the significance of the input descriptors on the target responses (feature engineering). SHapley Additive exPlanations (SHAP) was analyzed to study the importance of the features on CO conversion and C_{5+} selectivity based on the game theory. One-way and two-way partial dependence plots (PDPs) were implemented using scikit-learn library to shed light on the possible dependencies between the catalyst performance descriptors (activity and selectivity) and a set of input features, while keeping other features at their mean values. The overall steps of the proposed ML framework are exhibited in Fig. 3.

3. Results and discussion

3.1. Analysis of input data

The collected dataset was statistically investigated to understand the distribution of the data in the dataset. The results of the boxplot analysis for some of the input features and responses are depicted in Fig. 4. The reaction temperature ranged from 200 to 350 °C, covering the range for both high-temperature and low-temperature FT reaction. The calcination temperature on the other hand contributed to a wider range from 200 to 700 °C with some outliers accounting for the carbon encapsulated metal nanoparticles (CEMNPs), and catalysts without calcination step, respectively [15,16]. Fig. 4b indicates the statistical distribution for textural properties of the catalysts and supports. The pore size of the catalyst ranged from 1.04 to 20.5 nm with mean value of 6.6 nm. Contribution of the carbon supports with diverse pore sizes (ranging from 0.9 to 19.4 nm) leads to developing a reliable ML model with acceptable generalization capability. Particle size of the catalyst is another critical factor included in the model, covered a diverse range (1.52–75.26 nm). The CO conversion as one of the responses, ranged from 0.3 to 99.1 % with mean value of 54.1 %, indicated the superior performance of the carbon-supported Fe and Co catalysts in FTS. Furthermore, C_{5+} selectivity with mean value of 53.1 % confirmed the high potential of the carbon supports compared to the refractory metal oxide supports for synthesis of the clean transportation fuels through FTS as also indicated in Table 3. Metal oxide supports due to the strong interaction with the active metal; negatively affect the reducibility of the active sites during FTS reaction and inhibits the chain growth probability [17].

The correlation between input features and output variables were investigated and the results are depicted in Fig. 5. To identify the possible correlations between the catalyst formulation descriptors and operational conditions, Spearman correlation analysis was used. The nonlinear correlations between the feature could not be evaluated using conventional linear correlations such as Pearson correlation analysis

[21]. The calculated correlations ranged from -1 to $+1$, indicating the positive or negative correlations between the factors. The support and catalyst pore sizes were positively correlated with each other and with the catalyst activity (CO conversion), since the larger pores of the support improve the dispersion of the active sites and overcomes the mass transfer limitations leading to higher CO conversions [22].

On other hand, there is a negative correlation between reaction temperature and C_{5+} selectivity since higher operating temperatures for Fe catalysts increases the selectivity to methane and short-chain hydrocarbons [23]. Moreover, Co catalysts are usually operated under lower temperatures for production of the linear hydrocarbons. Negative correlations were also observed between the promoters loading and C_{5+} selectivity. Although adding promoter to the structure of the catalyst positively affects the catalyst activity and selectivity, but higher promoter loadings possibly block the active sites, which decreases the chain growth probability.

Active metal loading on the other hand positively correlated with C_{5+} selectivity and CO conversion. Higher active metal loadings affect the turnover frequency (TOF) in FTS, and this is possible by employing carbonaceous supports with engineered pore sizes and surface chemistry to prevent the agglomeration of the metal nanoparticles. In other words, high contribution of the mesopores and presence of surface functional groups, which can anchor the metal nanoparticles prevent the agglomeration of the active sites supported on carbon materials with engineered physico-chemical features. The operational conditions of the catalyst synthesis such as calcination temperature and time were highly correlated with CO conversion, depicting the effects of catalyst preparation conditions on the structural and reaction performances of the catalyst. The negative relation between the calcination time and CO conversion arises from the fact that longer calcination times increases the metal-support interactions and metal particle size, which negatively affect the catalyst performance [24].

Principal component analysis (PCA) was also employed to investigate and visualize the information space in multiple dimensions. According to Fig. 5a, the first principal component accounted for about 40 % of the total variance existed in the dataset. Furthermore, over 80 % of the total variance can be extracted from the first four principal components. Promoter type and loading were included in the first PC. Second PC on the other hand, comprised H_2/CO ratio, calcination time, catalyst, and support surface area of the catalyst, indicating the significance of the catalyst formulation and operational conditions on the catalyst performance in FTS. Operational conditions for synthesis of the catalyst and FTS reaction were included in PC3 and 4.

The projection of the experimental data based on the first two PCs were also investigated to scrutinize the information space (Fig. 6). Most of the experimental data were highly clustered along a linear plane. These points can be assigned to the experimental results achieved by variations in the operational conditions. Moreover, some outliers in the information space were observed (Fig. 6b). The two separated clusters

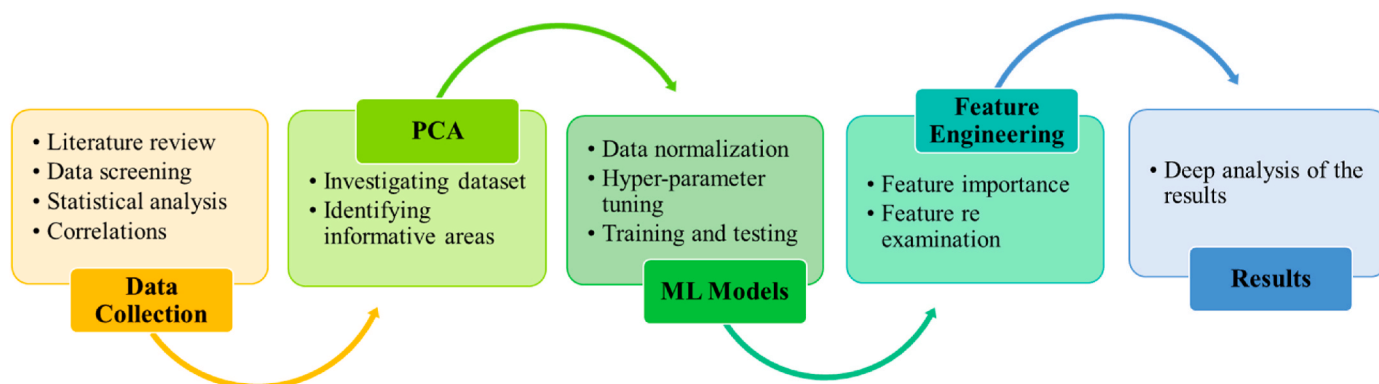


Fig. 3. The steps of the proposed ML framework.

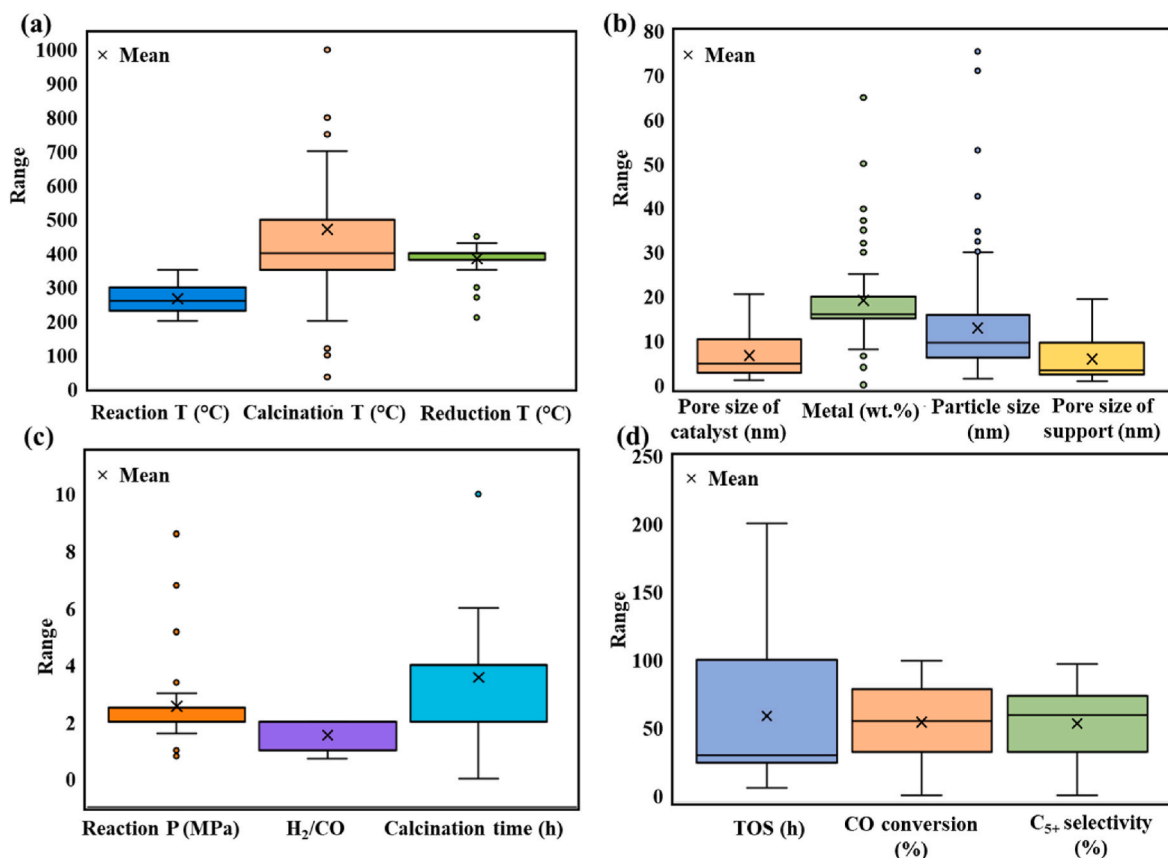


Fig. 4. Boxplot analysis of the a) reaction, calcination, reduction temperature, b) pore sizes of catalyst, catalyst support, active metal loading, particle size of the catalyst, c) reaction pressure, H_2/CO ratio of the syngas, calcination time, and d) time-on-stream, CO conversion, and C_{5+} selectivity.

Table 3

Comparison of the performance of the metal oxide and carbon supported catalysts for Fischer-Tropsch synthesis.

Catalyst	T (°C)	P (MPa)	GHSV (h^{-1})	CO conversion (%)	C_{5+} selectivity (%)	Reference
20Fe/ Al_2O_3	300	2	2000	30.8	64.6	[3]
20Fe/AC	300	2	2000	46.7	72.5	[3]
10Fe0.8K/ SiO_2	300	1	2200	16.9	62.8	[18]
10Fe0.8K/AC	300	1	2200	48.9	62.0	[18]
10Fe/ $\gamma-Al_2O_3$	300	1	2200	10.0	58.6	[18]
10Fe/ SiO_2	300	1	2200	16.7	45.5	[18]
10Fe/AC	300	1	2200	58.5	60.2	[18]
15Co/CNT	230	2	2400	48.3	61.7	[19]
$Fe_xO_y@C$	270	2	2000	2.5 (L/g-Fe.h)	60	[20]
Fe/ SiO_2	270	2	2000	1.6 (L/g-Fe.h)	44	[20]

identified in Fig. 6b can be related to the experimental data points with unique catalyst formulations (preparation methods such as mechanical mixing and employing novel carbon supports such as chitosan). Chitosan as a carbonaceous polymer with versatile functional groups and high-thermal stability can be a suitable support material for designing an effective FTS catalyst [25]. Presence of different functional groups on the surface of chitosan reduces the necessity of support pre-treatment before metal loading, which facilitates the catalyst synthesis process. The experimental data points with higher catalyst and support surface area (Fig. 6d, c) can be assigned to the catalysts supported on amorphous carbon materials such as AC and carbon black. The hierarchical pore structure in these materials, on one hand, ensures the uniform dispersion of the metal nanoparticles, on the other hand, improves the diffusion rate of the reactant molecules to the surface of the active sites by providing a suitable surface area. From the projection of the experimental data based on first two PCs, it is clear that some areas in the information space have not been explored in the literature, highlighting

the necessity of investigating novel catalyst formulations for FTS.

3.2. Optimization of the machine learning models

Three machine learning models (SVR, RF, and ANN) were evaluated and their hyper-parameters were tuned to effectively predict the CO conversion and C_{5+} selectivity of the FTS catalyst supported on carbon materials with different formulations under different operational conditions. Fig. 7 represents the results of the hyper-parameter optimization, predictions of the CO conversion (Y_1), and C_{5+} selectivity (Y_2) using SVR model with radial basis function (rbf) as kernel function. Epsilon and the regularization parameter as two significant variables of the SVR model were optimized for two responses and the results are shown in Fig. 7a, d. For Y_1 , the RMSE values were increased for epsilon values smaller than 0.1, but a downward trend was observed for RMSE values by increasing the epsilon values. The minimum RMSE was achieved with epsilon of 0.25 and regularization parameter of 4 for CO

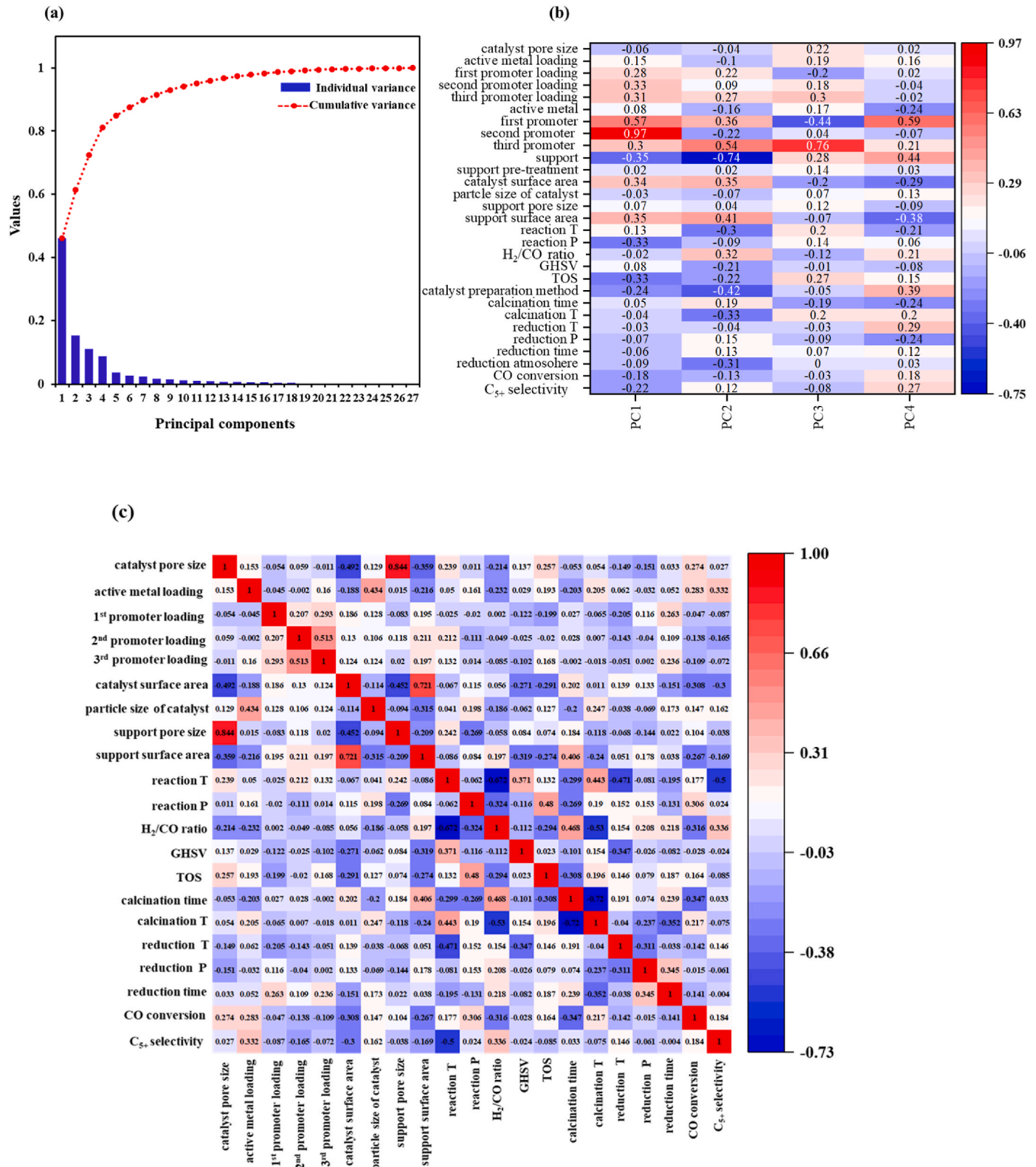


Fig. 5. Principal component analysis of the dataset: a) variance of each component, b) correlation of the input features and responses with the top-four principal components, and c) Spearman correlation matrix between the input features and output responses.

conversion. According to Fig. 7a, the RMSE values reached its minimum in epsilon, and regularization parameters smaller than 0.5 and 30, respectively for CO conversion. Epsilon value of 1, and regularization value of 81, on the other hand led to the minimum RMSE for C₅₊ selectivity (Fig. 7d).

RF as an ensemble machine learning model, combines several randomized decision trees and aggregates their predictions by averaging [3]. The prediction accuracy of the RF is affected by number of trees and the maximum depth. Fig. 8 depicts the results of the hyper-parameter optimization, predictions of the CO conversion (Y₁), and C₅₊

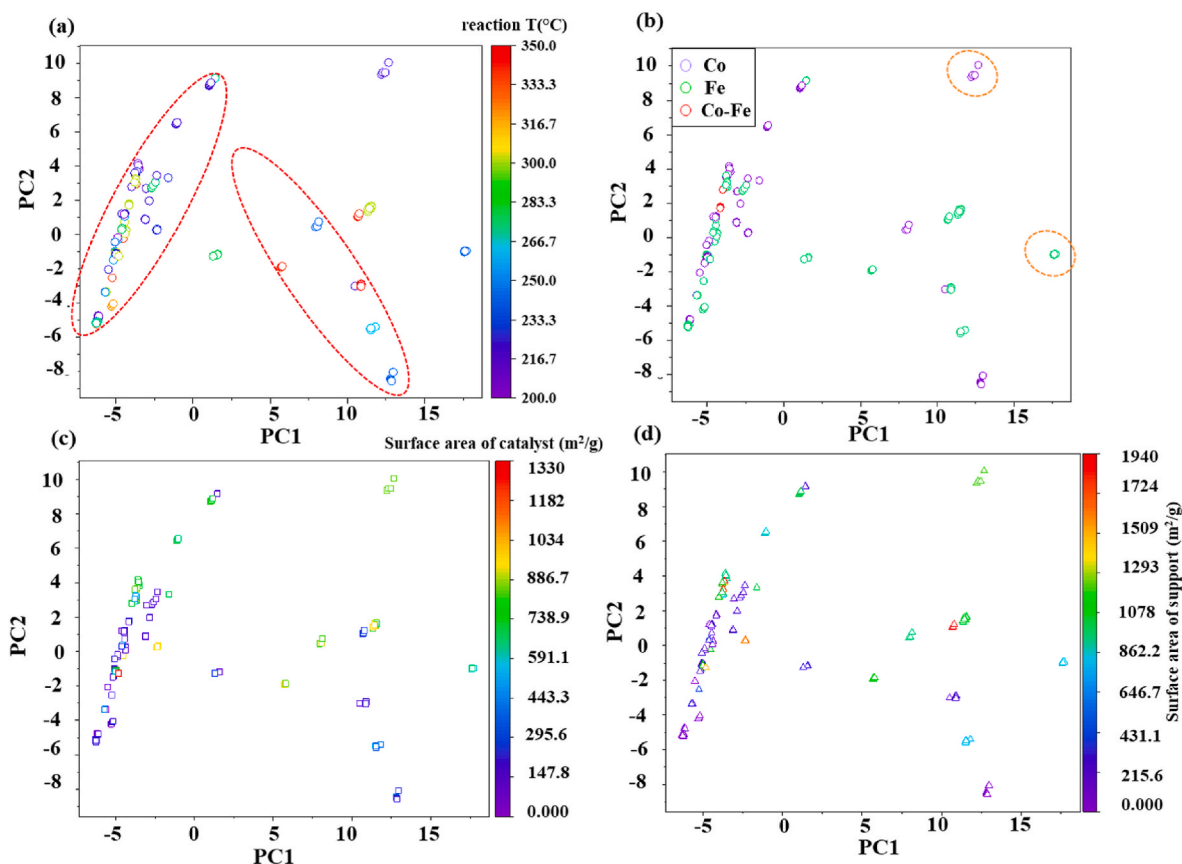


Fig. 6. PCs categorized by (a) reaction T, (b) type of active metal, (c) surface area of catalyst, and (d) support.

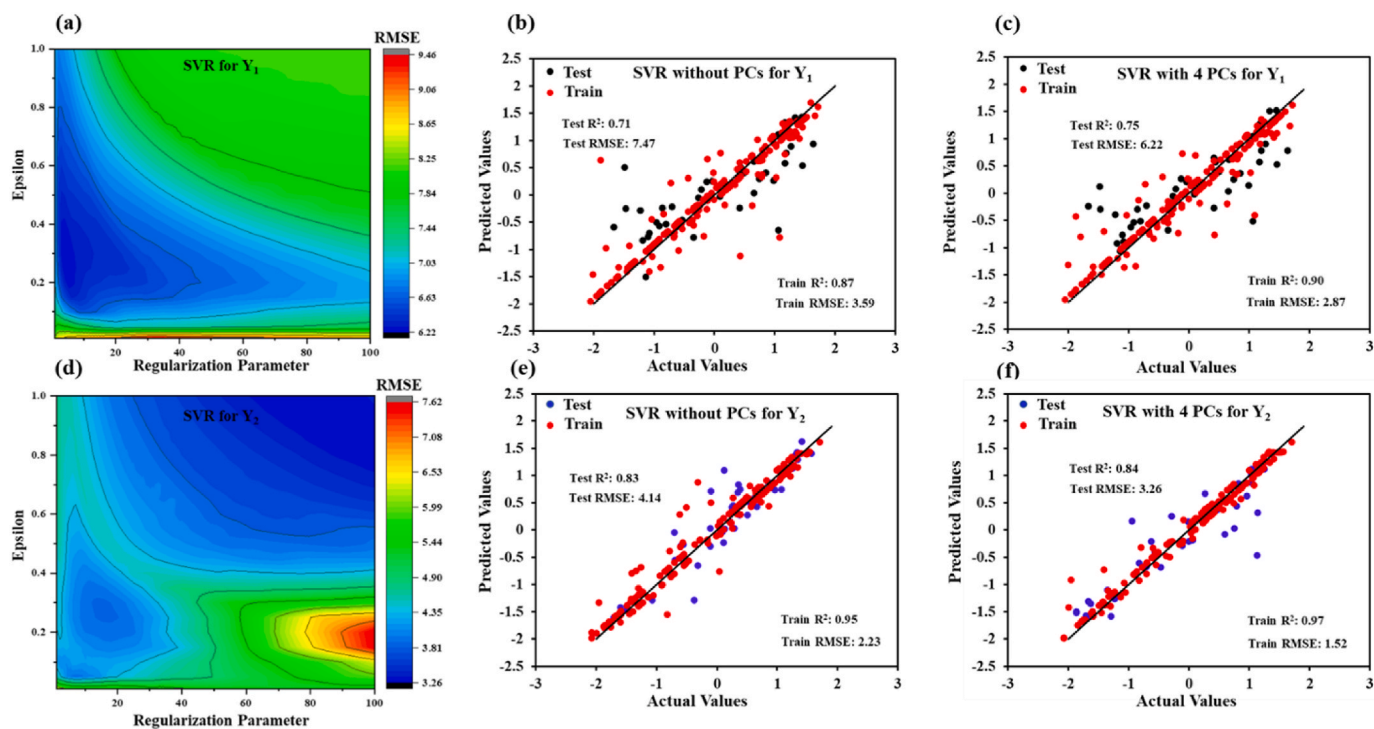


Fig. 7. (a) Hyper-parameter tuning of SVR for CO conversion, (b), (c) predicted data versus the original experimental data with and without PCs for CO conversion, (d) Hyper-parameter tuning of SVR for C₅₊ selectivity, (e), and (f) predicted data versus the original experimental data with and without PCs for C₅₊ selectivity.

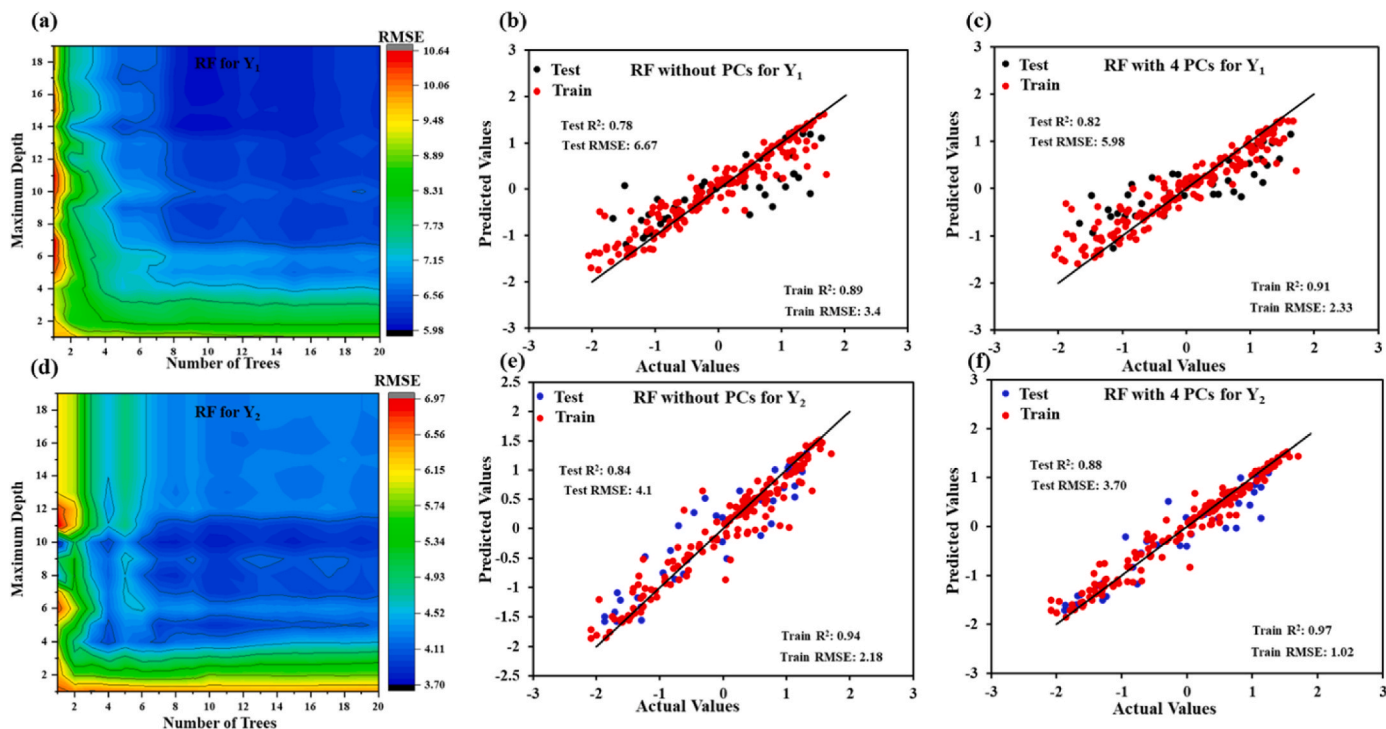


Fig. 8. (a) Hyper-parameter tuning of RF for CO conversion, (b), (c) predicted data versus the original experimental data with and without PCs for CO conversion, (d) Hyper-parameter tuning of RF for C_{5+} selectivity, (e), and (f) predicted data versus the original experimental data with and without PCs for C_{5+} selectivity.

selectivity (Y_2) using RF model. For the first response (Y_1), minimum RMSE was achieved using 9 trees with maximum depth of 14 (Fig. 8a). On the other hand, for Y_2 (C_{5+} selectivity) maximum depth of 10 and 8 trees minimized the RMSE (Fig. 8d).

The structure of the ANN network used in this work was optimized in terms of number of neurons in the first and second hidden layers (the schematic of the ANN model is shown in Fig. S2 (supplementary

material)). As depicted in Figs. 9a and 16 neurons in the first and 9 neurons in the second hidden layer led to the minimum RMSE of 6.51 for CO conversion as the first target (Y_1). For C_{5+} selectivity (Y_2) the minimum RMSE of 4.89 was achieved using 15 and 9 nodes in the first and second hidden layers, respectively (Fig. 9d).

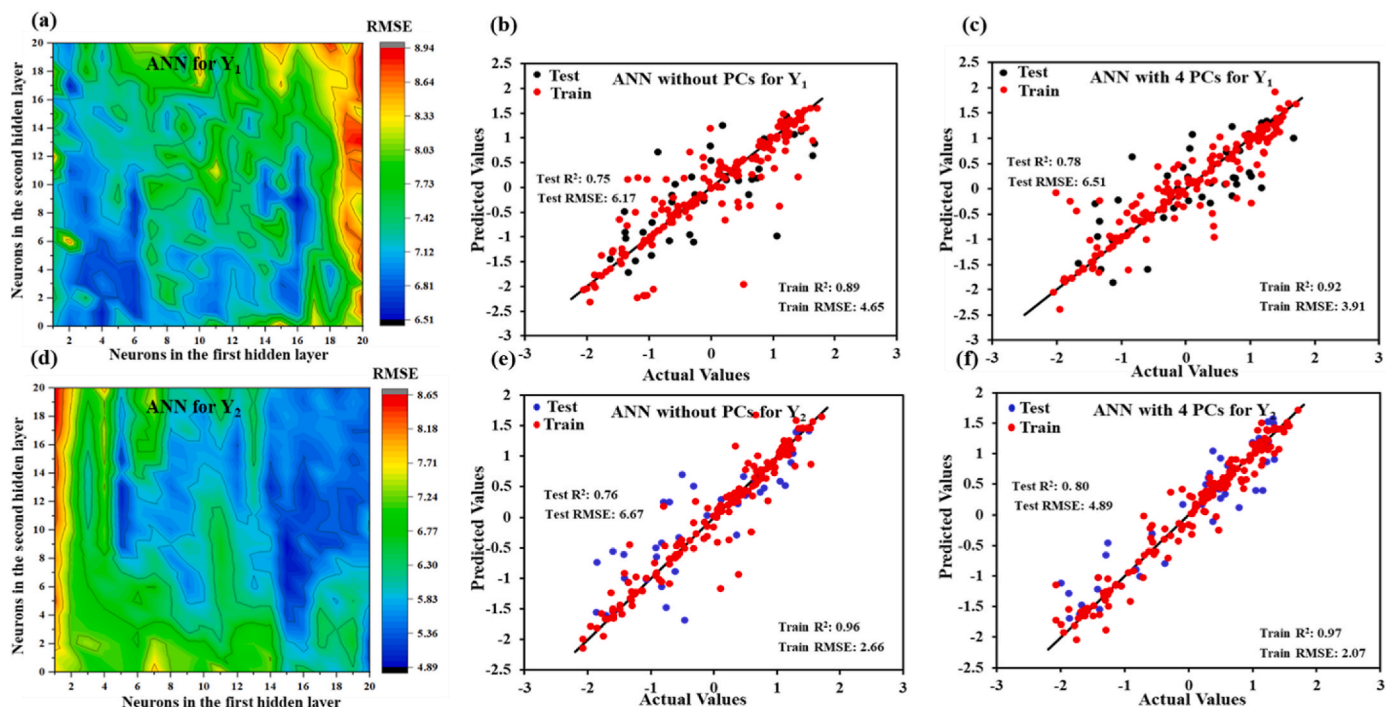


Fig. 9. (a) Hyper-parameter tuning of ANN for CO conversion, (b), (c) predicted data versus the original experimental data with and without PCs for CO conversion, (d) Hyper-parameter tuning of ANN for C_{5+} selectivity, (e), and (f) predicted data versus the original experimental data with and without PCs for C_{5+} selectivity.

3.3. Model evaluation

After optimization of the machine learning models (section 3.2.), their prediction capabilities were evaluated. The prediction robustness of the models was analyzed using two scenarios: considering PCs and without PCs. Evaluation of the ML models considering the PCs, helps us to capture the FTS performance of the Fe and Co catalysts supported on carbonaceous materials using a dimensionally reduced dataset. Considering PCs, simplifies the ML modeling by reducing the time is required for model training. In other words, PCA improves the prediction ability of ML models, because it identifies the areas that are sparse or dense in the dataset [26]. The experimental data versus the predicted values of the CO conversion (Y_1) and C_{5+} selectivity (Y_2) is presented in the form of scatter plots in Figs. 7, 8, and 9 using SVR, RF, and ANN methods, respectively.

It is evident from Fig. 8, that the data points for training and testing for both CO conversion and C_{5+} selectivity responses were denser using RF model compared to the SVR and ANN, highlighting the higher interpretability of RF model. In terms of accuracy, the average R^2 values for Y_1 based on RF model were reported as 0.91, and 0.82 for train and test data, respectively. On the other hand, for Y_2 , the average R^2 values for train and test data were reported as 0.97 and 0.88, respectively. For both responses R^2 values for train and test data were larger using RF model, suggesting the higher prediction accuracy of this model.

Although the prediction accuracy of the three ML models (SVR, RF, and ANN) were lower in test dataset compared to the train data; but the prediction performance of all three models was sufficient and comparable to exhibit a good generalization of FTS activity and selectivity for catalysts supported on carbon materials. RF outperformed SVR and ANN for prediction of the CO conversion and C_{5+} selectivity in the FTS using

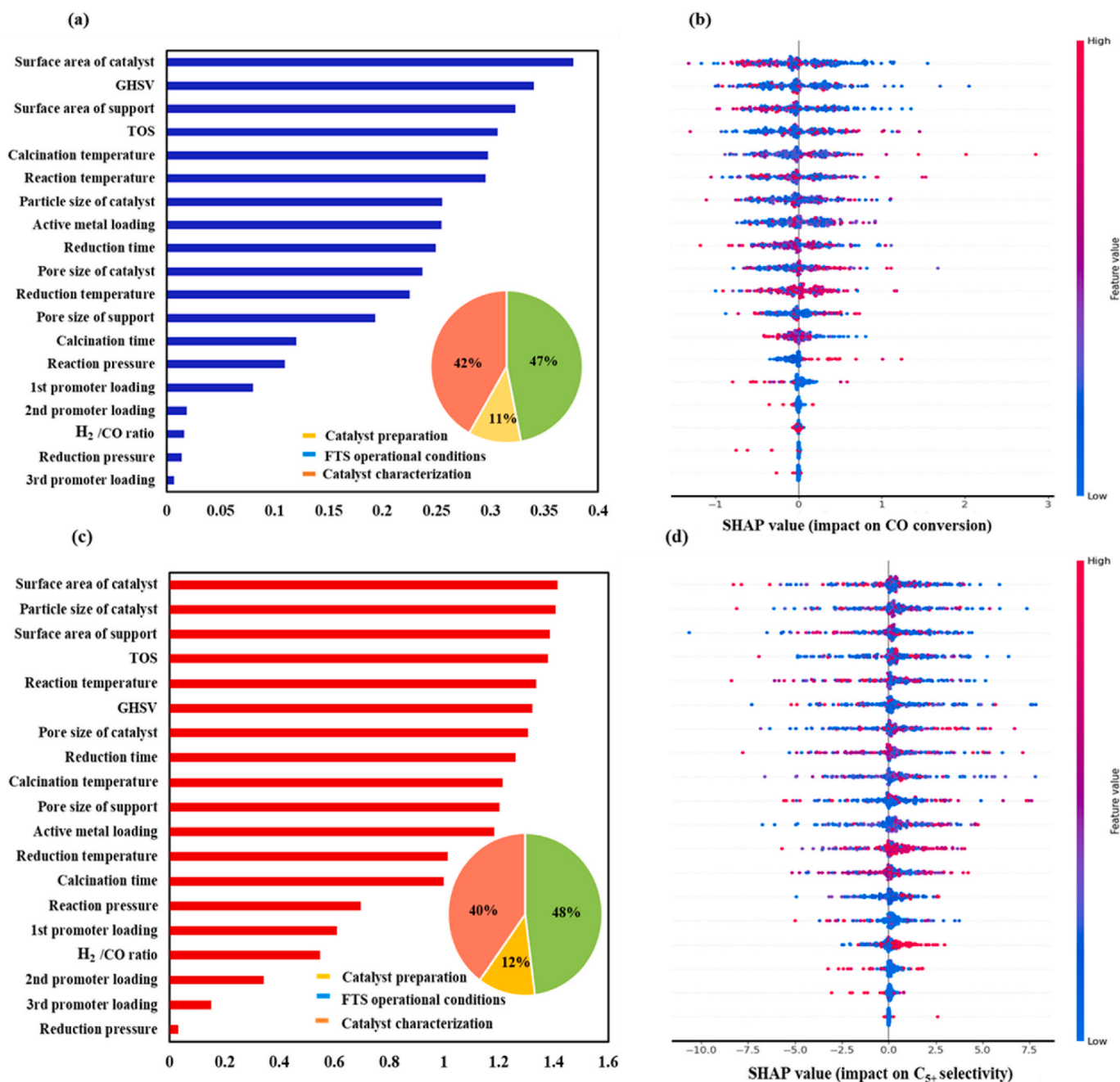


Fig. 10. Impacts of input features on (a), (b) CO conversion, (c) and (d) C_{5+} selectivity based on RF model using SHAP values.

Fe/Co catalysts supported on carbonaceous structures.

Including qualitative input features such as support pre-treatment, catalyst preparation method, and support type in the ML model built in this work helps us to reduce the bias that might arise from the focus on the numerical data and improve the model interpretability. Moreover, considering the effects of the qualitative features in the dataset improve the prediction accuracy of the models, and can shed light on the complex reaction mechanism and products distribution in FTS by using catalysts with specific formulations. The ML model was also validated with 3 new datasets and the results depicted satisfactory prediction accuracy for CO conversion and C₅₊ selectivity (Table S1, supplementary material).

3.4. Impacts of the input features on catalyst activity and selectivity

The importance of the input features on CO conversion and C₅₊ selectivity of the catalysts supported on carbon materials was elucidated using SHAP theory. This analysis was only carried out for RF model, which outperformed ANN and SVR as discussed in section 3.3. Fig. 10 indicates the impacts of input features, categorized as FTS operational conditions, catalyst characterization, and preparation, on CO conversion and C₅₊ selectivity as bee swarm plots. SHAP values (Fig. 10b, d) represent the contribution of each input data on the outputs. While, the blue color depicts the lower original value for the input feature, red points show the higher original values for the input descriptors [27]. According to Fig. 10b, surface area of the catalyst, support, GHSV, TOS, calcination, reduction, FTS reaction temperatures, time, pore size of catalyst, support, particle size of catalyst, and active metal loading were identified as important parameters for prediction of the CO conversion. Fig. 10a, c also presents the bar and pie charts of the mean absolute SHAP values of the CO conversion and C₅₊ selectivity, respectively.

Surface area of the catalyst and support indicated a negative effect on both CO conversion and C₅₊ selectivity (color of the data points were changed from blue to red with the decrease of SHAP value). Carbonaceous supports with higher surface area such as AC consist of higher contribution of the micropores, leading to catalysts with relatively high surface area. It is reported that metal particle sizes in the range of 10–20 nm leads to the efficient performance of the carbon-supported catalysts in FT reaction [28]. On the other hand, pore size of the support presented a positive effect on CO conversion and C₅₊ selectivity of the FT catalysts supported on carbon materials, highlighting the critical role of mesopores in the structure of the carbon supports (3). Moreover, active metal loading also showed a positive effect on catalytic activity and selectivity to heavier hydrocarbons; higher metal loading ensures more active sites for hydrogenation of CO to long-chain hydrocarbons [29]. On the other hand, higher metal loadings lead to possible metal agglomeration, which can be counter-balanced by the optimum metal-support interactions, and relatively high surface area with higher contribution of the mesopores in the structure of the carbon supports. Catalyst preparation conditions (calcination temperature and time) were identified as other significant features on the catalytic activity and selectivity. While calcination temperature presented a positive effect on CO conversion, an obvious positive or negative effect of the calcination temperature on the C₅₊ selectivity was not detected. Changing calcination temperature directly affects the structural and textural characteristics of the catalyst [30].

FTS operational conditions, which consisted of both catalyst activation and FTS conditions had marked effects on the catalytic activity and selectivity. GHSV was one of the critical FTS operational condition parameters presenting a negative contribution in prediction of the CO conversion (Fig. 10a and b). Higher GHSV values, decreases the residence time of syngas in the reactor, negatively affecting the CO conversion and chain growth. To improve the catalytic activity and selectivity to long-chain hydrocarbons, the reactants should be supplied to the reactor before desorption of the CH₄ from the surface of the catalyst, to ensure the chain propagation [31]. FTS reaction temperature is another controlling factor in predicting the FTS performance using

Fe/Co catalysts supported on carbon materials. While reaction temperature manifested a positive influence on CO conversion, a negative trend was observed for C₅₊ selectivity. Increasing reaction temperature has a profound effect on CO conversion, especially for Fe catalysts. On other hand Co catalyst are operated at lower temperatures, leading to higher selectivity to heavier hydrocarbons. It has been reported that increased reaction temperature shifts the selectivity of the products to light olefins instead on C₅₊ hydrocarbons for Fe/Co catalyst supported on carbon materials [32]. Further increasing reaction temperature has been reported to be detrimental on the catalytic performance by promoting the Boudouard reaction (the main reason of catalyst deactivation through carbon deposition) [33]. This negative correlation between the reaction temperature and C₅₊ selectivity was also observed in Spearman correlation matrix (Fig. 5c).

Catalyst activation is a key step in improving the performance of the FTS using Fe and Co catalysts as indicated in Fig. 10. While Fe carbides are assumed the active phases for Fe catalysts, metallic Co is responsible for hydrogenation of CO using Co-based catalysts [34]. The efficiency of the catalyst activation depends on the reduction time, temperature, and atmosphere, which control the structure of the catalyst during FTS reaction. Application of the carbon encapsulated metal nanoparticles (CEMNP) as FTS catalysts, improves the catalyst activation. Metal carbides are partially present in the structure of the CEMNPs, which facilitates the deoxygenation in the activation and reaction steps [35]. According to the mean absolute SHAP values (Fig. 10a, c), catalyst characterizations and FTS operational conditions were the dominant factors affecting the CO conversion and C₅₊ selectivity of the catalysts supported on carbon materials.

Some of the features such as promoters loading, H₂/CO ratio, and reduction pressure indicated insignificant impacts on the products distribution and catalytic activity of FTS. Therefore, to decrease the computational cost of the employed ML framework, raised from the redundant features, re-examination of the input features was carried out. Four features (2nd, 3rd promoter loadings, H₂/CO ratio, and reduction pressure) were removed for prediction of CO conversion, and two parameters (3rd promoter loading and reduction pressure) were removed for C₅₊ selectivity. The updated results after feature re-examination using RF model are summarized in Table 4. The optimized hyper-parameters discussed in section 3.2. were used for RF model training and testing with updated features. The prediction accuracy of the RF model using updated features (feature re-examination), was improved for CO conversion, and marginally decreased for C₅₊ selectivity. The results show that updating the number of input features based on the SHAP values led to useful insights on engineering the feature selection to improve the prediction performance of the ML models and optimize the computational process.

To further analyze the marginal effect of a single feature on the CO conversion and C₅₊ selectivity of the Co-Fe catalysts supported on carbon materials, one-way PDPs were studied (Fig. 11). The tick marks in the x axis indicate data density. The influence of catalyst surface area on CO conversion was increased within the range below ~350 m²/g; but further increase in the surface area of the catalyst not only did not improve the catalyst activity, but also slightly decreased CO conversion (see Fig. 11a). Higher surface area of the catalyst sheds light on dominant contribution of the micropores in the structure of the carbon

Table 4
Comparison of the prediction accuracy of the RF model with updated features.

Response	Test-R ² with PC		Test-RMSE with PC	
	All features	Updated features	All features	Updated features
CO conversion	0.82	0.89	5.98	4.12
C ₅₊ selectivity	0.88	0.86	3.70	3.78

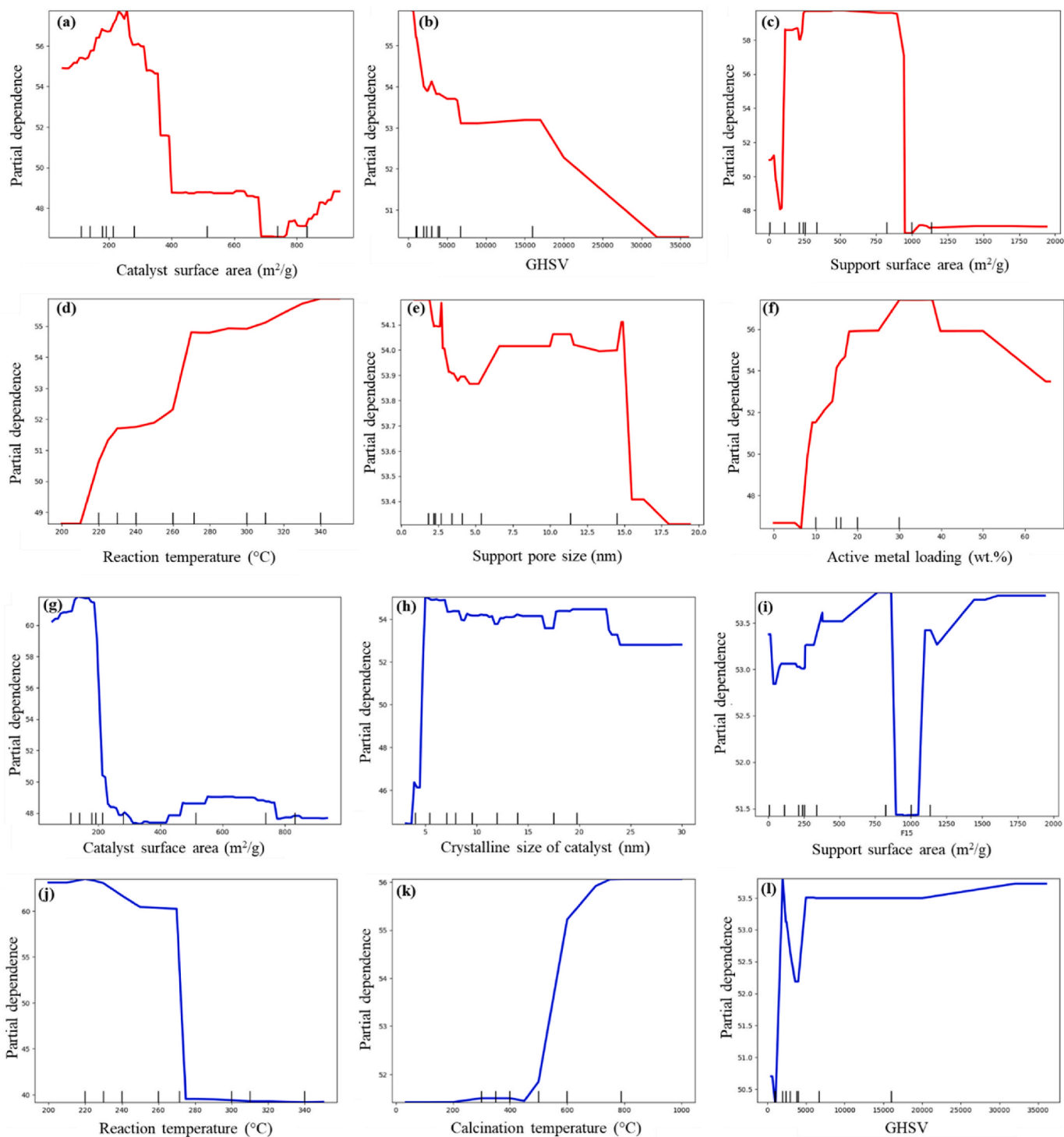


Fig. 11. One-way partial dependence plots (PDPs) of CO conversion (a–f), and C_{5+} selectivity (g–l) on some of the important catalyst formulation and operational variables.

support, which lead to possible agglomeration of the active sites [3]. GHSV is another significant factor on predicting the catalytic performance of the carbon supported Fe/Co catalysts in FTS (confirmed by SHAP results). CO conversion continuously decreased by increasing the GHSV; which confirms the noticeable impact of the optimum residence time for syngas molecules in the reactor on the catalyst performance. GHSV values greater than 5000 h^{-1} substantially decreased CO conversion (Fig. 11b).

Partial dependence of the CO conversion on the support surface area

is depicted in Fig. 11c. The catalysts with approximately $0 \text{ m}^2/\text{g}$ surface area of the support imply the contribution of the core-shell catalysts in the dataset. Carbon supports with surface area in the range of $250\text{--}850 \text{ m}^2/\text{g}$ demonstrated the highest CO conversion, corresponding to the carbon supports with higher contribution of the mesopores such as CNT [36]. Although micropore-dominant catalyst supports such as AC demonstrated larger surface areas; but negatively affected the catalyst performance in FTS, which can be originated from the non-uniform dispersion of the active sites intensifying mass transfer resistance [37].

Reaction temperature and active metal loading are other major determinants of the catalyst performance modifying the activation energy of the FTS, and number of active sites, respectively. Positive correlations were observed between the CO conversion, reaction temperature, and active metal loading (Fig. 11d, f).

The PDP analysis confirmed that C_{5+} selectivity substantially dropped as catalyst surface area increased, but the dependency remained unchanged for surface areas $>300 \text{ m}^2/\text{g}$ (see Fig. 11g). Catalysts with optimum surface area ($\sim 300 \text{ m}^2/\text{g}$) contribute to the carbon supports with well-defined mesopores with crystallized graphitic structure, which positively affected the active metal dispersion, carburization, and chain growth to heavier hydrocarbons. An exponential rise was observed in the partial dependency of the C_{5+} selectivity with the crystal size in the range of 2–7 nm, further highlighting the structure sensitivity of the FTS reaction [38]. Catalysts with crystal sizes in the range of 10–20 nm depicted a consistent C_{5+} selectivity (see Fig. 11h).

Reaction temperature on the other hand, demonstrated an inverse relation with the selectivity of the heavier hydrocarbons (see Fig. 11j). Increasing reaction temperature promotes the reaction towards

production of the CH_4 , and shorter chain hydrocarbons. Higher C_{5+} selectivity of the catalyst supported on carbon materials at lower operational temperatures can be assigned to the Co catalysts, which are more applicable for low-temperature FT (LTFT) process [39]. Relatively improved C_{5+} selectivity of the Fe/Co catalyst with carbon supports even at higher reaction temperatures; can be originated from the surface pre-treatment of the carbon supports, which increases the electron density on the active sites, improving the CO dissociation and C–C coupling [40]. Calcination temperature in the range of 200–400 °C did not significantly affect the C_{5+} selectivity, but a monotonic trend was detected for calcination temperatures $>450 \text{ °C}$ (see Fig. 11k). Improved C_{5+} selectivity of the catalysts at higher calcination temperatures are related to the carbon encapsulated metal nanoparticles (CEMNP), or iron/cobalt-metal organic frameworks (MOF) as catalysts for FTS. CEMNP and MOF due the controlled crystal size of the active phase, higher graphitization degree and improved carburization in the case of Fe catalyst; indicate improved selectivity to liquid hydrocarbons compared to other carbon supported catalysts [41].

The two-way partial dependence plots of CO conversion and C_{5+}

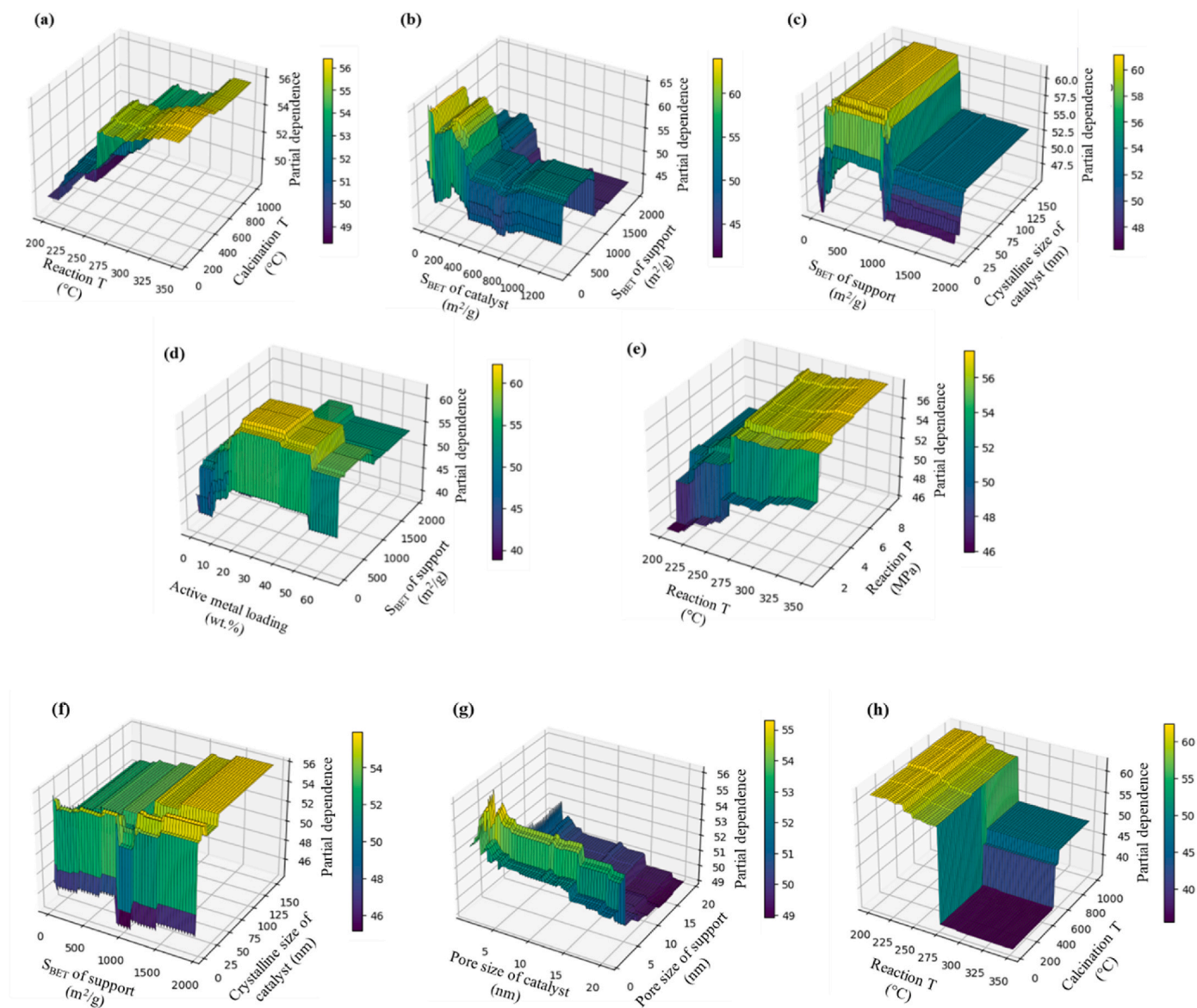


Fig. 12. Two-way partial dependence plots (PDP) of CO conversion (a–e) and C_{5+} selectivity (f–h) on any two important input variables and the interactions between them. The surfaces are colored according to their relative CO conversion and C_{5+} selectivity values (blue: low, and yellow: high). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

selectivity is represented in Fig. 12. The partial dependence was high with calcination temperatures in the range of 400–800 °C, and reaction temperatures 300–325 °C (see Fig. 12a). The synergetic effects of calcination and reaction temperatures were also observed in the Spearman correlation matrix (Fig. 5). The high interactions and non-linear relation between catalyst, support surface areas, and catalyst activity were also validated in the two-way PDPs (Fig. 12b).

Carbon supports with BET surface area in the range of 500–1000 m²/g, depicted the highest CO conversion with corresponding catalyst surface areas of 100–400 m²/g. The negative interaction between BET surface of the support and the crystal size of the catalyst, which was initially revealed with Spearman correlation matrix (Fig. 5), further validated by PDPs (Fig. 12c). Although dominant contribution of the micropores in the structure of the carbon support improves surface area, active metal dispersion; but negatively affects the CO conversion by mass transfer limitations in the presence of relatively small crystal size of the catalyst. While smaller Co particles sizes inhibits H₂ dissociation; smaller Fe nanoparticles exhibit lower chain growth probability [42]. A monotonic relation between the crystal size of the carbon supported catalysts and selectivity to heavier hydrocarbons was observed in two-way PDP (Fig. 12f). On the other hand, support material with surface area in the range of 500–1000 m²/g positively contributed to the chain growth in FTS. The higher CO hydrogenation ability of the Co catalyst compared to Fe catalyst is responsible for the higher selectivity of C₅₊ hydrocarbons at lower operational conditions (see Fig. 12h) [43].

These plots together demonstrate the preference for an optimum catalyst nanoparticle size dominates the determination of the catalyst activity and selectivity to C₅₊ hydrocarbons. Furthermore, to design a catalyst with optimum nanoparticle size, employing carbon support with engineered textural properties and surface chemistry is of high significance. Calcination temperature also highly affected the CO conversion and C₅₊ selectivity of the carbon supported Fe/Co catalysts based on the one-way and two-way PDPs, by modifying the graphitic structure of the carbon supports, and active site-support electronical interactions.

4. Conclusions

Herein, a machine learning framework was developed to explore the performance of the carbon supported Fe/Co catalysts in FTS to produce clean liquid fuels as a sustainable energy resource. The dataset consists of 27 input features and two final targets was compiled from the literature. Before ML modeling, the dataset was statistically analyzed, which showed that the dataset covered a wide range of the data (various catalyst formulations and operating conditions) leading to prediction with acceptable generalization capability.

PCA was studied to explore the information space and it was found out that over 80 % of the total variance can be extracted from the first four principal components. According to the projection of the experimental data based on the first two PCs, majority of the data was clustered around a line arising from modification of the operating conditions. The possible nonlinear correlations between the input features were analyzed using Spearman correlation matrix. A very strong positive correlation was detected between the pore size of the catalyst, support material, and the catalyst performance, assigning to the fact that support material with larger pore sizes improve the dispersion of the active sites improving the chain growth.

The ML models were trained and tested using two strategies: with PCs, and without PCs. All three models performed well in presence of PCs, highlighting the significance of exploring dataset before modeling. Using the optimized hyper-parameters, RF with RMSE of 5.98 and 3.70 for CO conversion and C₅₊ selectivity, respectively, outperformed ANN and SVR. The significance of the input features on the prediction capability of RF model was investigated using SHAP theory, which revealed that the promoter loading, reduction pressure, and H₂/CO ratio were at least importance for prediction of the catalytic activity and selectivity.

After removing the redundant input parameters and re-examination of the features, the prediction capability of the RF model for CO conversion was improved from R² of 0.82–0.89.

Partial dependence plots (PDPs) were also investigated to interpret the performance of the ML framework on predicting the FTS catalytic performance, which is influenced by catalyst formulation and operating conditions descriptors. PDP results revealed the coupled effects of textural properties of the support material, catalyst, operational conditions for catalysts synthesis and FTS reaction on the efficiency of the FTS process. This work can provide useful information for designing novel and cost-effective carbon-based supports for FTS catalysts to improve the selectivity of the liquid fuels. Sparsely traversed regions of the presented model can also be resolved by considering the effects of other descriptors such as electronic or structural fingerprints of the catalyst in the dataset.

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CRediT authorship contribution statement

Zahra Teimouri: Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Writing - original draft. **Nicolas Abatzoglou:** Supervision, Writing - review & editing. **Ajay K. Dalai:** Funding acquisition, Project administration, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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